

MATH 114 HW 4

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Question 1

a. Define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\mathbb{E}(X) = \mu$ since $X_0 \sim \pi, X_i \sim \pi \forall i$. We have

$$\begin{aligned}\mathbb{E}(\bar{X}) &= \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) \\ &= \frac{1}{n} \cdot n\mathbb{E}(X) \\ &= \mu\end{aligned}$$

b. We have $\mathbb{P}(X_{t+s} = j, X_t = i) = \mathbb{P}(X_{t+s=j}|X_t=i)\mathbb{P}(X_t = i)$

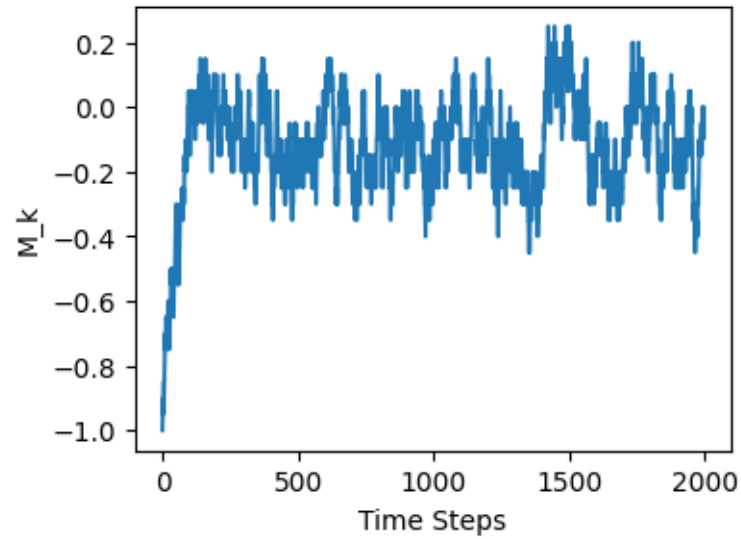
$$\begin{aligned}
\mathbb{E}(X_t X_{t+s}) &= \sum_{i,j=1}^N (ij) \mathbb{P}(X_{t+s} = j, X_t = i) \\
&= \sum_{i,j=1}^N (ij) \mathbb{P}(X_{t+s} = j | X_t = i) \mathbb{P}(X_t = i) \\
&= \sum_{i,j=1}^N (ij) (P_{ij})^s \cdot \pi(i) \\
&= \sum_{i,j=1}^N (ij) \mathbb{P}(X_{0+s} = j | X_0 = i) \mathbb{P}(X_0 = i) \\
&= \sum_{i,j=1}^N (ij) \mathbb{P}(X_s = j, X_0 = i) \\
&= \mathbb{E}(X_0 X_s)
\end{aligned}$$

c. $\text{Var} \bar{X} \neq \frac{1}{n} \text{Var} X$ because X_i depends on X_{i-1} , therefore they are not independent, and we can not compute the sum $\sum_{i=1}^n \text{Var} X_i$.

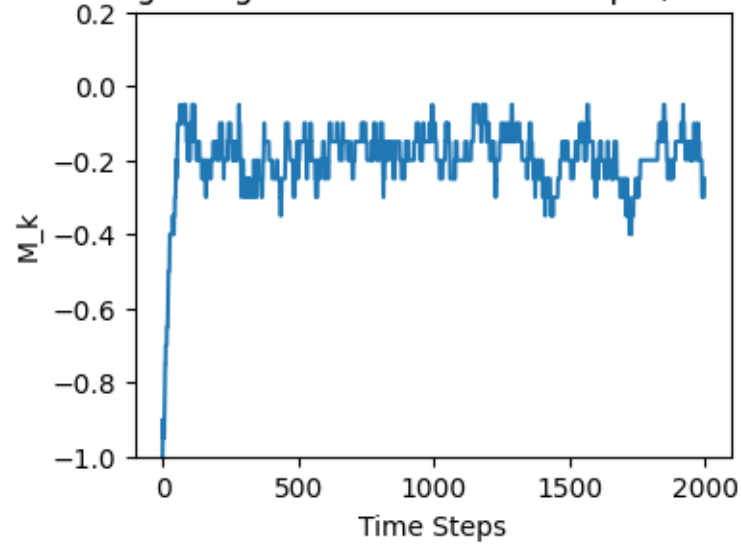
Question 2

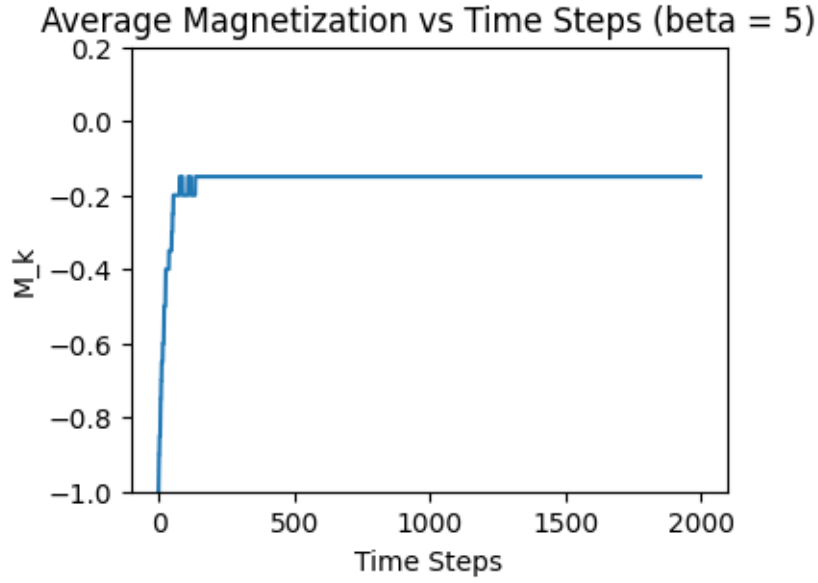
a. i.

Average Magnetization vs Time Steps (beta = 0.1)



Average Magnetization vs Time Steps (beta = 1)



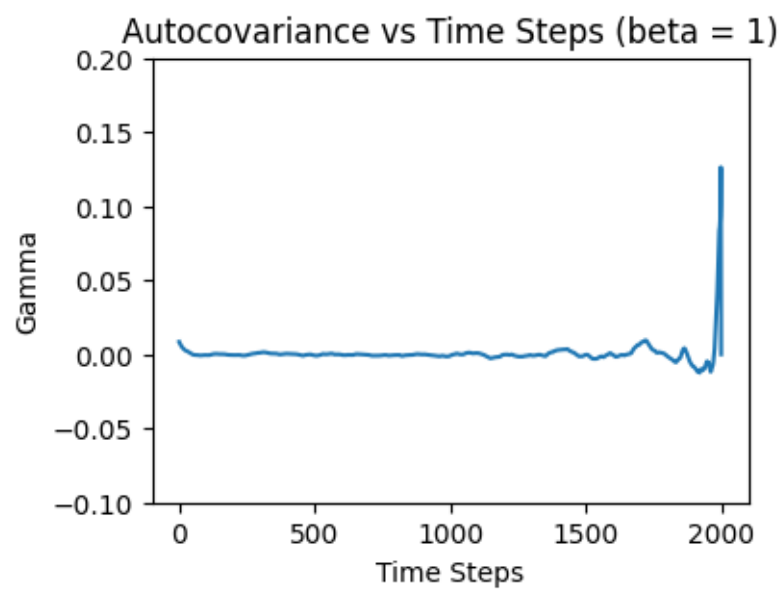
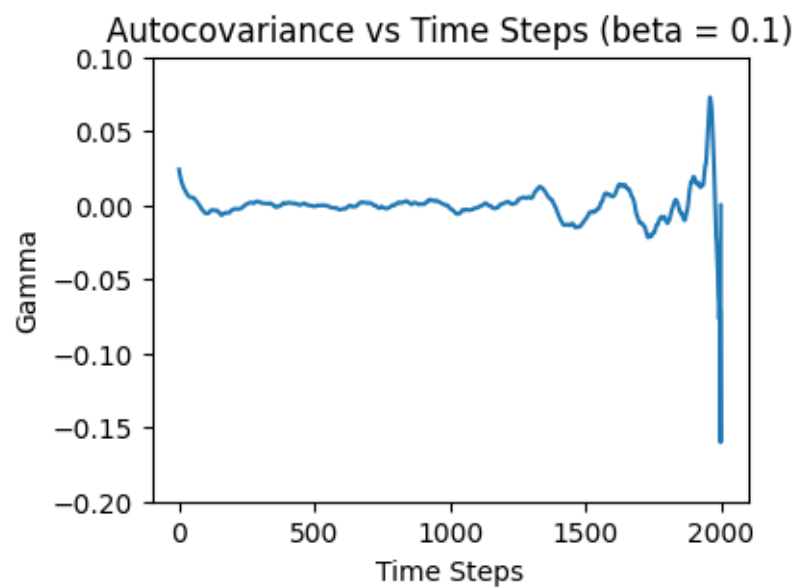


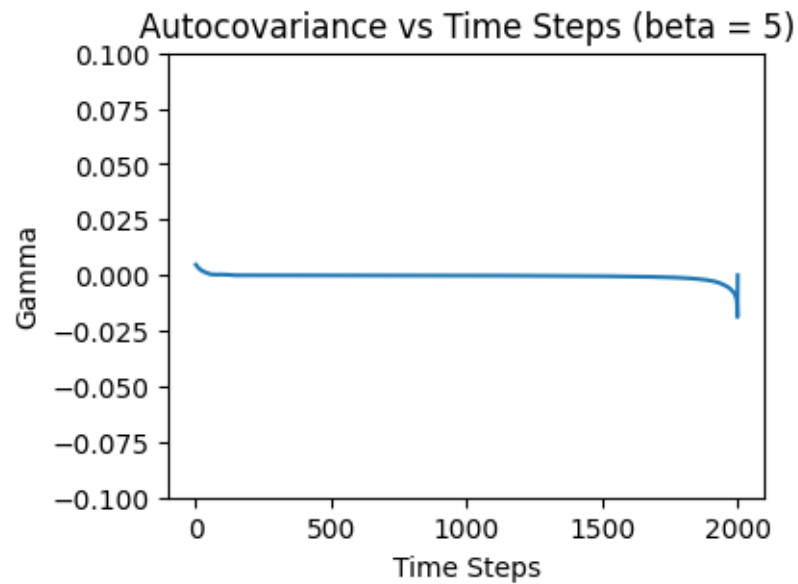
Note that

$$\alpha = \frac{f(W)}{f(X_i)} = \frac{\exp(-\beta H(W))}{\exp(-\beta H(X_i))} = \exp(-\beta(H(W) - H(X_i)))$$

Therefore the bigger the β , the lower α , and the more likely we accept our proposal state and the less that our average magnetization will fluctuate. This matches what we see within the graphs, where the higher the β 's, the less the graph fluctuates around -0.15 .

ii.





Autocovariance seems to decrease slightly as we added more time steps to the simulation, however there is a weird spike at the end of the simulation around 2000 time steps.

b. skipped

Question 3

skipped